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Preparation and characterization of novel polyols-based DESs and their use in efficient sequestration of radioactive iodine

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ABSTRACT

The utilization of Novel DESs can be greener alternative to traditional organic solvents. In present study, four polyols-based DESs are synthesized and characterized by several techniques. The prepared DESs are used for the efficient capture of hazardous iodine. In present study by simple combination of two readily accessible, economical, and biodegradable components four polyols-based DESs have been prepared. For the creation of these DESs, a hydrogen bond donors (HBDs) (polyols) and hydrogen bond acceptors (HBAs) (BTEAC) are used, which have melting points far lower than their respective melting points. The developed DESs were investigated for their physicochemical properties such as density, dynamic viscosity. Our results revealed distinct trends in these properties across different DES compositions, highlighting unique molecular interactions and solvation capabilities. The FTIR and NMR characterization study indicated considerable intermolecular interaction between the various component of the DES system. Further, Kamlet-Taft solvatochromic parameters were determined using Nile red dye, and other solvatochromic dyes. For synthesized DESs, the solvatochromic parameters E_T^N, normalized polarity parameter, π*, polarity / polarizability; β, hydrogen-bond acceptor basicity and α, hydrogen-bond donor acidity, have been determined. The E_T^N parameters demonstrate that both non-specific and specific interactions play a significant role in solute–solvent interactions. The synthesized DESs have been utilized for the efficient capture of the iodine. The removal efficiency of DES depends upon the mass of the DESs taken and time duration. These findings deepen our understanding of DES behavior and underscore their versatility in diverse industrial and scientific applications.

1. Introduction

Deep eutectic solvents (DESs) are a new class of solvents those have gained significant attention in recent years due to their unique properties and potential applications [1,2]. DESs are formed by mixing two or more components, typically a hydrogen bond donor and a hydrogen bond acceptor, by forming hydrogen bonds in DESs governs their unique properties and functionality, influencing solvation behavior, solvent structure, and overall performance) [3,4]. which results in the formation of a eutectic mixture with a lower melting point than either of the individual components. This low melting point and the unique properties such as high thermal stability [5], low viscosity [6,7] and high solubility that make them attractive media for various applications [7]. They are non-toxic, biodegradable, and have low volatility, which makes them environmentally friendly. The properties of high melting point/viscosity/polarity of DESs can be further tuned by varying the components and their ratios and render them suitable for various applications [8–10]. Over the past ten years, nuclear reactors have consistently supplied 10 % or less of the world's energy. Nuclear energy is anticipated to become more significant since it is a reliable and low-carbon energy source. Safety issues, nevertheless, continue to hinder its functioning. The volatile radioactive waste generated during the reprocessing of spent nuclear fuels—which essentially comprises radionuclides like ^{129}I and ^{131}I in the form of molecular iodine (I_2), etc. [11,12]. This is one of the biggest safety concerns. For the efficient removal and storage of these radioactive elements, DESs play a novel role in this era of chemistry [13–18]. Because of their advantageous qualities, such as low cost, simple preparation, and environmentally safe starting materials, deep eutectic solvents (DESs) have attracted a lot of attention as prospective extracting agents in recent years. Iodine excess is becoming more common, raising worries about the risks associated with it, including hypothyroidism, hyperthyroidism, malignancies, autoimmune thyroid

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disease (ATD), etc. [19]. Today, it is understood that iodine excess poses a risk to both human and animal development [20]. A significant area of worry is the rise in the amount of radioactive iodine and its isotopes in the environment. Iodine's involvement in metabolic processes and biomagnification caused radioactive iodine to accumulate in both the human body and at every level of the food chain. It has become a frightening situation for scientists and researchers to collect and remove radioactive iodine from the environment because of the thyroid gland's direct involvement with iodine in the human body. Through exposure to contaminated air or water, it can build up in the thyroid gland of both humans and animals. This can raise the risk of leukemia, thyroid cancer, and metabolic abnormalities. Due to its unusually long half-life of 1.57 years, ^{129}I is detrimental to the biosphere. While ^{131}I has a relatively short half-life of 8.02 days, when it builds up in human thyroid tissue, it can raise the risk of thyroid cancer. It is clear from past experiences that radioactive iodine causes thyroid damage and may be carcinogenic when it builds up in the atmosphere as a result of any radioactive fallout. In this study, developed four novel DESs are used for I_2 removal and storage [21,22]. Utilizing I_2 in cyclohexane solutions and ultraviolet–visible (UV/vis) spectroscopy, the effectiveness of DESs at removing I_2 was investigated. The capture of iodine by DESs involves the formation of a stable complex between the iodine molecule and the solvent components. In the present study, novel DESs have been prepared hydrogen acceptor species Benzyl triethyl ammonium chloride (BTEAC) and hydrogen bond donor species – e.g., Resorcinol (RES), Catechol (CH), Phenol (PH), Pyrogallol (PY), in the molar ratio of 1:2. This DES formulation aligns with Type I DES classification (The HBA can be a quaternary ammonium salt, while the HBD can be a range of compounds such as carboxylic acids, amides, or alcohols), known for its eutectic behavior and versatile applications. The synthesized DESs are characterized by several techniques such as NMR, FTIR, and the structure of synthesized DES is confirmed with data FTIR and NMR spectroscopy [23,24].The viscosity and density of pure and modified mixtures also studied to give an understanding of possible interactions present in the system. Further solvatochromic parameters such as the hydrogen-bond acidity, hydrogen-bond basicity, and dipolarity/polarizability of synthesized DES made of ammonium-based salts as hydrogen bond acceptors (HBAs) and phenolic compound as hydrogen bond donors (HBDs) were determined in this work [25]. It furnishes the meaningful insight how the polarity of the DES components was affected by the chemical structure of the DES components. This study lays the groundwork for future research into the enhancement and practical implementation of this good potential solvent technology [22,26–29]. The synthesized DESs have been used for the efficient used of hazardous I_2. Overall, deep eutectic solvents are a fascinating and promising area of research that has the potential to revolutionize many different industries [30,31]. While there are still many challenges to be overcome, the unique properties of DES make them an exciting avenue for future innovation and discovery [26,32]. The future of deep eutectic applications is promising [33–35] with potential uses in various industries and the potential to contribute to the advancement of sustainable and green technologies [21,36–41]. While DESs have shown promise in various industrial applications, their suitability for therapeutic use remains a topic of exploration. Understanding their biocompatibility, toxicity profile, and potential interactions is essential for evaluating their viability in therapeutic applications [42–44].

2. Experiment

2.1. Materials

Benzyl triethyl ammonium chloride (98 %), resorcinol (99 %), catechol (99 %), pyrogallol (99 %), and phenol (99 %) were purchased from Sigma-Aldrich, USA, in the highest purity possible. Cyclohexane (98 %) and iodine (≥95 %) were purchased from Acros Organics, and all the materials were used without further purification [45,46].

2.2. Preparation of DESs

DESs were synthesized by mixing benzyl triethyl ammonium chloride (BTEAC) and phenolic compound (resorcinol, catechol, pyrogallol, and phenol) at a defined molar ratio (1:2) (Fig. 1).The continuous and stirring while heating to 80 °C for 1 h at air pressure result a stable homogeneous liquid. All of the prepared DESs were dried for up to three hours at 60 °C in a vacuum oven. The synthesized DESs were maintained in a desiccator and kept in sealed laboratory vials [47–49]. The composition of HBA and HBDs are shown in Table S1. The water content of the DES constituents was measured using a Metrohm 831 Karl Fischer coulometer that was taken into consideration during the preparation of each mixture. The water content for DES1, DES2, DES3 and DES4 are 2550, 1860, 2165, and 1540 ppm, respectively.

3. Methods

3.1. Solubility measurements

To check whether DESs are hydrophobic or hydrophilic in nature solubility test have been performed with water, benzene, and cyclohexane solvents (Table S2) [50,51].

3.2. Density measurements

The densities (ρ) of pure DESs and their mixtures with water (10 %, 25 %, and 50 % water) were measured using a U-tube digital density meter (Laboratory Density Meter (model no: DMA 5000 M), Anton Paar GmbH) with a repeatability of 1.105 g/cm^3 and 0.01 K, at atmospheric pressure. The observations were made at atmospheric pressure and a temperature range of 293–343 K with a temperature interval of 5 K. A built-in Peltier thermostat (PT-100) with an accuracy of ±0.01 K controlled the temperature measurement range (from 293 to 343 K). The density was determined with 1.4 10^{-6} g cm^{-3} as the standard deviation [52].

3.3. Viscosity measurements

Using the Microviscometer (model number: LOVIS 2000 ME) Microviscometer Module, Anton Paar GmbH, the viscosity of pure DESs and their mixture was measured at a temperature range from 293.15 to 343.15 K with a temperature interval of 5 K. A built-in Peltier thermostat with a precision of 0.01 K was used to regulate the measuring temperature (which ranged from 293.15 to 343.15 K). A syringe was used to transfer the necessary amount of liquid (around 1 ml), taking extra care to avoid introducing bubbles into the tube to prevent errors in the measurements. In this range of viscosity values the diameter of the capillary was 1.59 mm, and the error for this capillary was 0.01 % [52].

3.4. Fourier −transform infrared spectroscopy (FTIR) measurements

FTIR spectra of pure compounds and DESs were recorded by Perkin Elmer FTIR (8000) spectrophotometer at 298.15 K. The range of the scan was from 4000 to 400 cm^{-1} [53–56].

3.5. NMR measurements

NMR spectra of DESs have been recorded by JNM-ECZL series FT NMR in chloroform D and DMSO-d_6 solvents at 25 °C [55,57,58].

3.6. UV–vis spectroscopy and fluorescence-spectroscopy

The UV–Vis spectra of the stock solution of iodine in cyclohexane and in DES were measured using a UV2550 ultraviolet spectrophotometer (Shimadzu, Japan). Fluorescence spectra of dye in DES are recorded on Hitachi F-2700 FL Spectrophotometer, fluorescence spectra for

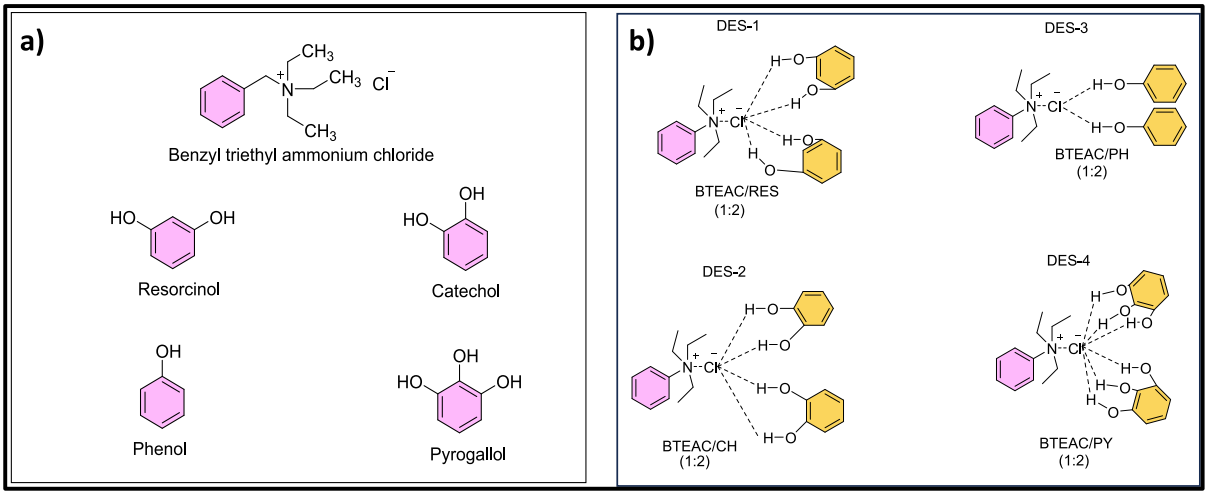


Fig. 1. (a) The structure of hba (bteac) and hbd (phenolic group) are shown. (b) The chemical structures of four des components are shown.

emission and excitation were captured using a quartz cuvette with a 1 cm path length and a 150 W Xenon lamp at at 298.15 K.

4. Iodine capture experiment

100 mL of cyclohexane was mixed with 0.5 g of solid iodine to create the iodine cyclohexane solution. The mixture above was then agitated to confirm the complete dissolution of stable iodine in cyclohexane.

The bottom of the quartz cuvette was then filled with 3 mL of the sample solution and 0.5 g of DES (aside from to explore the impact of mass). The spectrophotometer was used to measure the absorbance in situ at the absorbance maximum at 523 nm using a sealed quartz cuvette. It was also possible to measure the pure iodine-cyclohexane solution's absorbance. Data were gathered at the following intervals of time: 1 min, 5 min, 10 min, 0.5 h, 1 h, 1.5 h, 2 h, 3 h, 4 h, 5 h, 6 h, and 7 h [45]. (vide supra I_2 capture and kinetic).

5. Result and discussion

5.1. Structural properties of DESs

5.1.1. NMR and FTIR

The NMR investigation has been done for synthesized DESs (Fig. S2). The signal at $\delta = 9.00$ ppm attributed to the proton attached to hydroxyl group, further signal at $\delta = 9.35, 9.11, 9.09$, and 8.96 observed due to the proton attached to H, N, O group in (Fig. S2) shows the interaction of BTEAC with polyols. Fig. 2(a) and (b) depict the FTIR spectra of a pure components and four synthesized DESs, respectively. The stretching vibration of polyols C—O was to the bands at 1220 cm^{-1}. The two absorption peaks between 1450 and 1600 cm^{-1} are due to the C═C stretching vibration in the phenyl ring. While the peaks of OH groups in resorcinol and pyrogallol were close to 3250 cm^{-1} and the distinctive hydroxyl peaks in phenol and catechol were at 3400 cm^{-1}. A sharp peak that was attributed to free hydroxyls seen in (Fig. 2(a)). A strong peak at 2950 cm^{-1} to 3000 cm^{-1} and towering peaks between 700 cm^{-1} and 790 cm^{-1} are characteristics of benzyl triethyl ammonium chloride (BTEAC). The four DESs' FTIR spectra differ significantly from one another (Fig. 2(b)). Because of the association of the hydroxyl group involving in hydrogen bonding, n(OH) in all DESs exhibited a broad peak at 3150 cm^{-1}. This broadening of the peak hints about how chlorine molecules are involved in hydrogen bond formation. The creation of the N—Cl bond was also indicated by the change in FTIR frequency of three different oscillation modes, namely at v_1, 3010 cm^{-1} to 3029 cm^{-1}; v_2, 1724 cm^{-1} to 1741 cm^{-1}; and v_3, 3156 cm^{-1} to 3124 cm^{-1}. It shows after the formation of the DESs, the red shifts and broadening of the O—H and C—H stretching peaks can be attributed to the formation of hydrogen bonds between BTEAC and polyols.

5.2. Physicochemical properties of DESs

The different physicochemical properties such as density, viscosity, and solubility, etc, are of utmost important and give valuable information regarding the structure of DES further, the application of DES depends on the physiochemical properties. Therefore, physicochemical properties of DESs including solubility, viscosity, and density, are studied extensively.

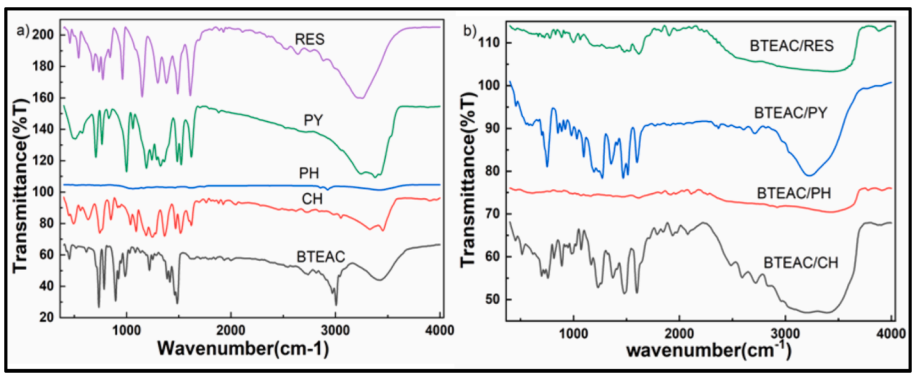


Fig. 2. (a) FTIR spectra of pure RES, PY, OH, CH, and BTEAC. (b) FTIR spectra of four prepared DESs.

5.2.1. Solubility

The solubility of the DESs are checked in different solvent schematically. Generally, it is reported that all synthesized DESs are hydrophilic in nature. Here it is found that DESs are utterly miscible in the water; (Fig. S3 and Table S2) [59]. And further the solubility of prepared DESs is also checked in different organic solvents. All four DESs were immiscible to cyclohexane and formed a layer.

5.2.2. Density

5.2.2.1. Density of the pure DESs and their mixture with water. The densities of all neat polyalcohol-based DESs' and their mixtures with water were measured at various temperatures. The higher density measured was (1.21 g cm^{-3}) for (BTEAC/PY) DES-4 is observed, while the lowest was (1.086 g cm^{-3}) reported for (BTEAC/PH) for DES-3 at a 10 % water dilution. Follow the order: DES-4 > DES-2 > DES-1 > DES-3. The variation of density with temperature is shown in Fig. 3. Density is an important physicochemical property of DESs which furnishes the important information regarding the molecular structure of the DES. The density of all synthesized DESs are measured at different temperatures and has been given in Fig. 3. The density of pure DES follows the order DES-4 > DES-1 > DES-2 > DES3 as shown in Fig. 4. Further, there is a decrease in density as increasing the temperature of the system. Further, the data were fitted according to equation $\sigma = \sigma_0 + aT$shown in Fig. 3. The recovered parameter has been given in Table S4.

5.2.2.2. Effect of water on DES density. Study of the density of DES is an important tool for comprehending the physicochemical properties of deep eutectic solvents. Fig. 5. depicts the variation of the DESs density with the weight % of water in DESs. The density of the DESs decreases as the weight percentage of water increases. Further data has been fitted with the equation with equation $\sigma = \sigma_0 + aW$, and recovered parameters are given in (Table 2). This is due to the fact that water is less dense than the DES. Further, a prominent decrease in the density of DES 2 is observed that those of other DESs (Table 1). It is reported that the density of DES depend on the molecular structure of DES and the packing within DES. It is hypothised that the addition of the water disturb the efficient packing of HBD and HBA component due to interaction with either one. Which result in decrease in the density [56]. It is important to mention that density of the DES can be adjusted for the

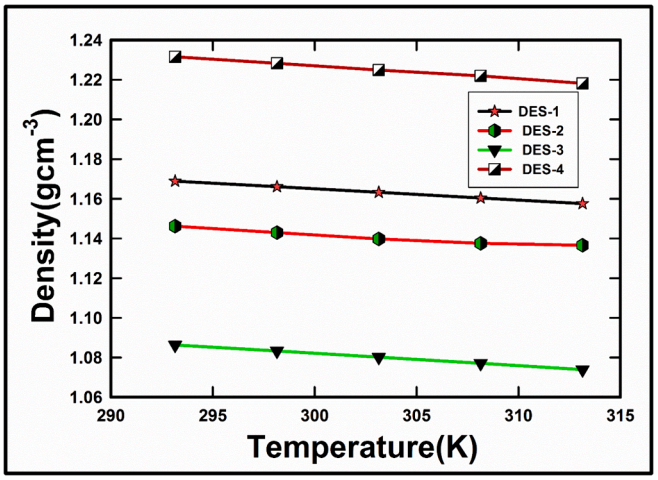


Fig. 4. Variation of density for four DESs (pure) as a function of temperature.

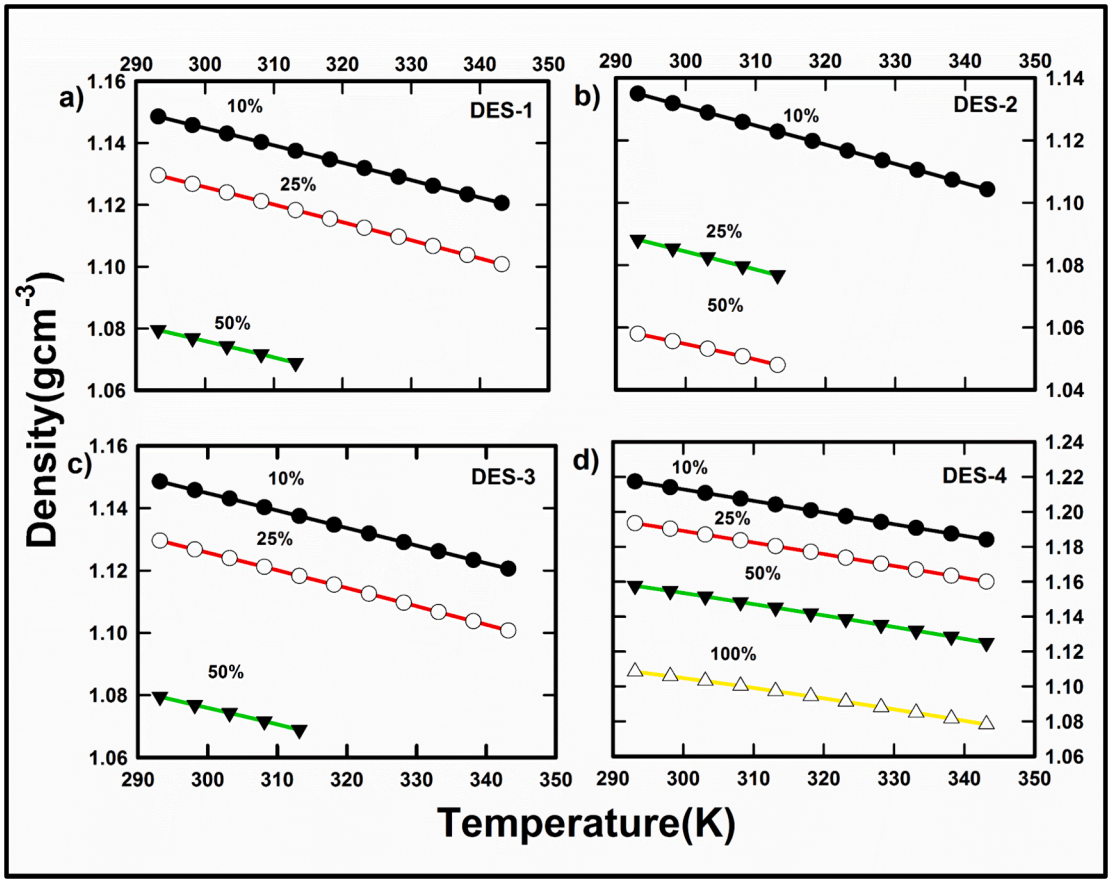


Fig. 3. Variation of density of synthesized DESs + water system (10 %,25 %,50 % and 100 %) where panel (a) represents DES-1, (b) DES-2, (c) DES-3, and (d) DES-4 as a function of temperature (290–350) K at 101.13 kPa at different dilutions. Solid lines give the fitting to data with equation $\sigma = \sigma_0 + aT$.

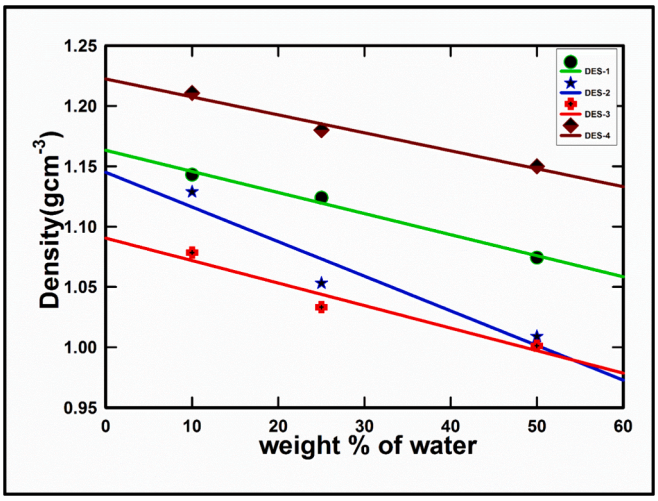


Fig. 5. Shows the variation in the density of DESs with different weight% of water and the solid line shows fitting by equation $\sigma = \sigma_0 + aW$. The recovered parameters are given in Table 2.

Table 1
Shows the variation of density (g cm^{-3}) of pure DESs at different temperatures.

Temperature (K)	Density (g cm^{-3})			
	DES1	DES2	DES3	DES4
293.15	1.1689	1.1462	1.0863	1.2316
298.15	1.1661	1.1430	1.0833	1.2283
303.15	1.1633	1.1398	1.0802	1.2249
308.15	1.1604	1.1376	1.0771	1.2220
313.15	1.1576	1.1366	1.0739	1.2182

Table 2
Shows the recovered parameters by fitting density of aqueous DESs by equation $\sigma = \sigma_0 + aW$ at 298 K and (101.13 kPa) pressure (Fig. 5).

DES	σ_0	a	S.D.
DES-1 (BTEAC/RES)	1.1633	−0.0017	0.0054
DES-2 (BTEAC/CH)	1.1450	−0.0029	0.0249
DES-3 (BTEAC/PH)	1.0905	−0.0056	0.0132
DES-4 (BTEAC/PY)	1.2224	−0.0015	0.0065

specific application by controlling the weight % of water.

5.3. Dynamic viscosity

5.3.1. Dynamic viscosity of the DESs and their mixture with water

The rate of diffusion of a DES is influenced by its viscosity, which is a crucial factor for different physicochemical process. Therefore the viscosities of synthesized DES has been determined at different temperatures and the results. (Table 4) shows lowest observed viscosity was 3.5591 kg m^{-1} s^{-1} (DES-3) at 293.15 k on 100 % dilution and the maximum measured viscosity was 4.3415 kg $m^{-1}S^{-1}$ (DES-4). Viscosity and temperature have a reverse exponential relationship. The sample's viscosity at ambient temperature is in the same order as its density: DES-4 > DES-2 > DES-1 > DES-3. Further, there is a decrease in viscosity as increasing the temperature of the system. Further, the data were fitted according to equation $\eta = \eta_0 e - E\eta/RT$ is shown in Figs. 6 and 7. Further, the comparison of the viscosity of four synthesized DESs of four DESs + water (10 %) is shown in Fig. S4. This might be because the HBD's long-chain moiety interacts closely with the HBA, increasing noncovalent interactions and the viscosity of the medium under investigation. The viscosity of the DESs appear to be significantly affected by temperature. Fig. 6 shows the variation of dynamic viscosity with temperature [21,60].

6. Effect of water on viscosity

The viscosity of DES is greatly influenced by the addition of cosolvent such as water, etc. Fig. 7. depicts the relationship between the DES's viscosity and the weight percentage of water in the DES. The viscosity of the DES decreases as the weight percentage of water in the solvent increases. This relationship exists because water has a low viscosity in comparison to the other components of the DES. If DES is to be used in a flow-through process, a lower viscosity may be preferred to facilitate solvent flow. It is possible to create a highly efficient deep eutectic solvent with a customized viscosity that satisfied the needs of particular application. Further, it is worthy to mention here, that addition of small amount of water to DES drastically decrease the viscosity of DES. On addition of 25 % of water, there is 100-fold decrease in viscosity. This behaviour of the viscosity of aqueous DES mixture may be due to large differences in the size and/or the shape of the component species of the mixture and the loss of Coulombic attractive interaction (or dipolar association) or rupture of H-bonding network within the mixture as compared to that in the neat components which result in sharp decrease in the viscosity. The viscosity data at different wt% of water is fitted by equation $\eta = \eta_0 e - E\eta/RT$ and the recovered parameters have been given in Table 3.

6.1. Kinematic viscosity

6.1.1. Temperature-related variation in DES kinematic viscosity

The kinematic viscosity (m^2s^{-1}) of the synthesized DESs is also measured. It measures a fluid's internal resistance to flow under gravitational force [59]. It is calculated by dividing dynamic viscosity by density at the same temperature. And it follows the same pattern as dynamic viscosity. The highest one is observed in DES-4, and the lowest one is observed in DES-3 and follows sequence DES-4 > DES-2 > DES-1 > DES-3. Fig. 8. and 9 show the variation of kinematic viscosity with temperature. Further, there is a decrease in kinematic viscosity as increasing the temperature of the system. Further, the data ln (kinematic viscosity) DESs + water (10 %) with1/T were fitted according to equation $\eta = \eta_0 e - E\eta/RT$ is shown in Fig. 8. The recovered parameter has been given in Table S7. Further, the comparison of the kinematic viscosity of four synthesized DESs of four DESs + water (10 %) is shown in Fig. S5 (see Fig. 9).

6.2. Fluorescence probes study of DESs: Nile red and coumarin-102

Nile red and coumarin-102 are among widely used photoinduced charge transfer (PCT) fluorescent probes. The major reason for the notable differences in their fluorescence responses of these probes of microsphere region. Specific solute–solvent interactions, changes in molecular structure, intersystem crossing to the triplet state, and monophotonic photoionization may also have an impact on their fluorescence responses in different media.[83] However, it is well known that the fluorescence emission of these two probes varies bathochromically as the polarity of their cybotactic zone is increased. Fig. 10. displays the Nile red and coumarin-102 fluorescence emission maxima for four polyols-based DESs. Fig. 10(a) shows the fluorescence emission spectra of Nile Red dissolved in four DESs. According to a detailed analysis of the data, DES-4's λ_{ex} max of Nile Red has the higher value out of the four DESs. Nile Red demonstrates that DES-4 is the most dipolar among the four DESs, the least dipolar, on the other hand, is found to be DES-3. These two DESs (DES-1 and DES-2) have identical dipolarity with λ_{ex} max 642. The characteristics of coumarin-102 make it a very effective fluorescence probe. Fig. 10(b). shows the fluorescence emission spectra of coumarin-102 dissolved in four DESs, and Fig. 10(b) reports the associated emission maxima (λ_{ex} max). Coumarin-102 DES-3 λ_{ex} max has the lowest value, indicating that it is the least dipolar DES among the

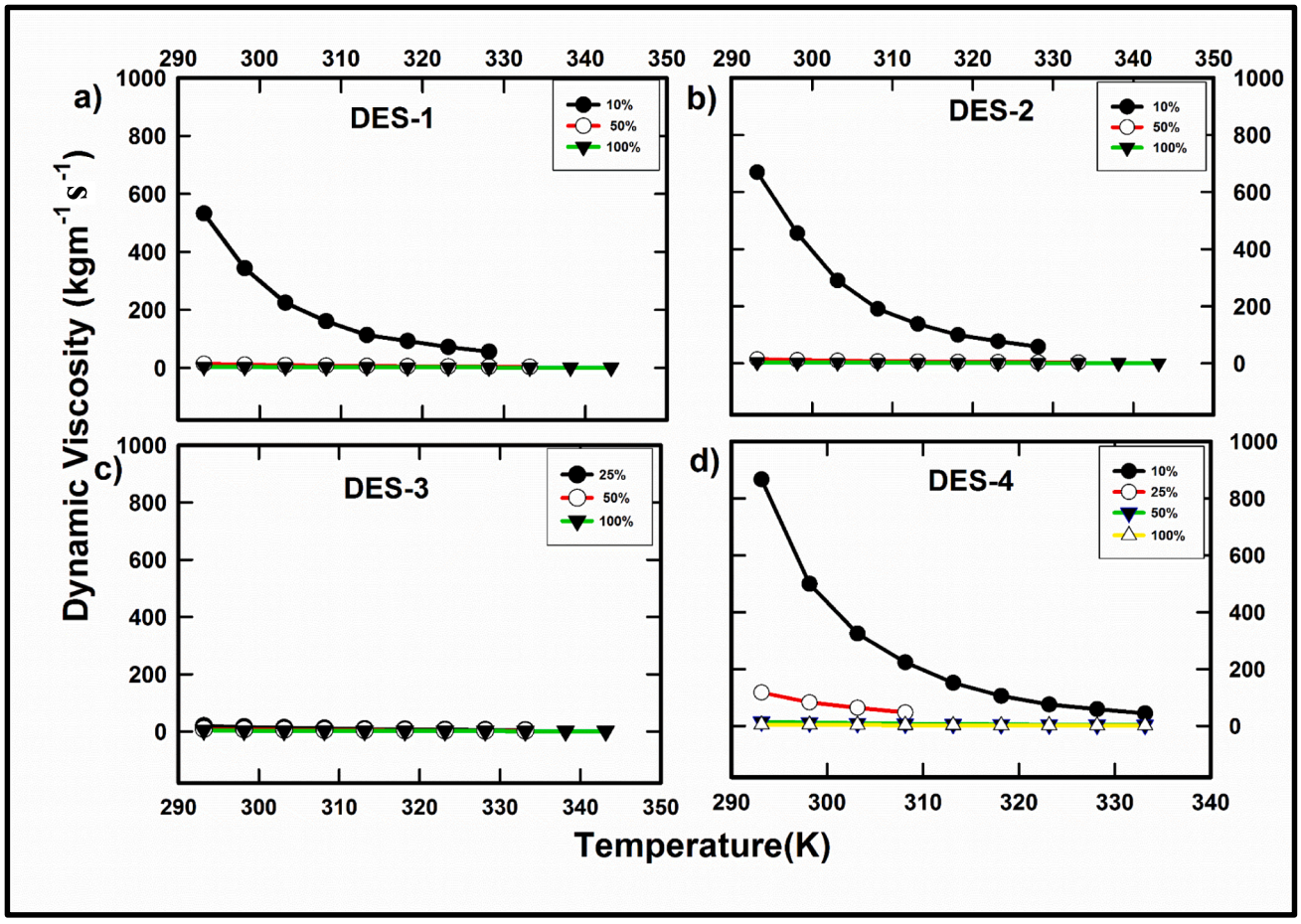


Fig. 6. Variation of dynamic viscosity (kg m^{-1} s^{-1}) for four synthesized DESs + water (10 %, 25 %, 50 %) as a function of temperature (290–350 K) at 101.13 kPa, where panel a represents DES1, panel b, c, d are for DES 2 DES3 and DES 4, respectively.

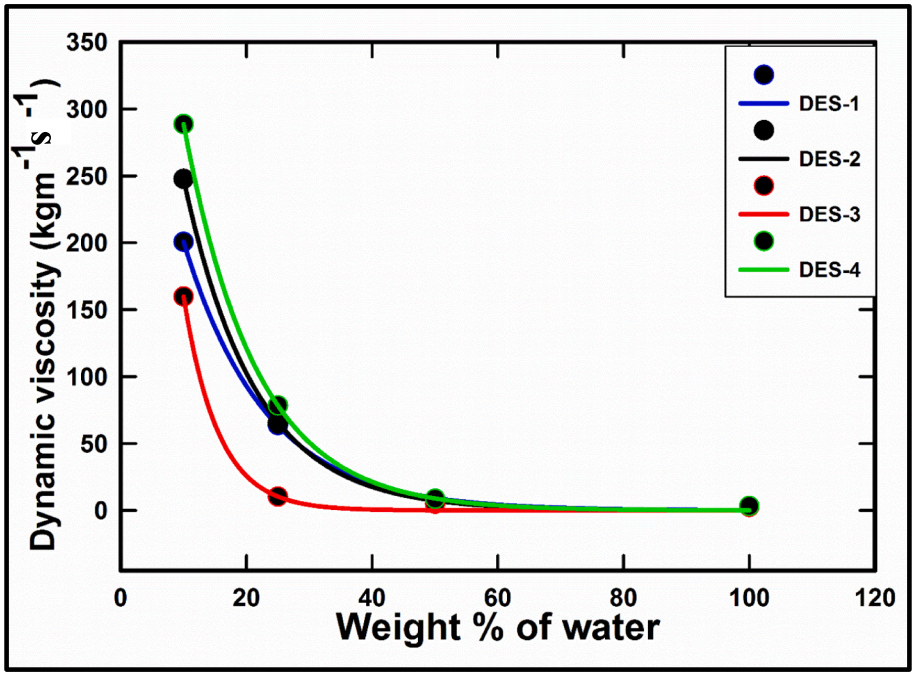


Fig. 7. Shows the variation of dynamic viscosity with different weight % of water and solid line shows data fitting by equation f = a*exp(−b*x).

Table 3
Shows the variation of dynamic viscosity with different temperatures and curve is fitted by equation f = a*exp(−b*x) (the standard error in viscosity was-3.3923 kg m^{-1} s^{-1}).

Temperature(K)	Dynamic Viscosity (kg m^{-1} s^{-1})			
	DES1	DES2	DES3	DES4
293.15	1315.4244	1730.7260	35.4565	3246.2471
298.15	801..6224	1127.2900	27.9430	1635.2779
303.15	494.6886	672.4270	22.2110	949.1990
308.15	340.4223	416.8017	18.0667	618.9559
313.15	228.2341	290.3175	14.8205	

Table 4
Shows the dynamic viscosity of pure DESs at 298 K (101.13 kPa) with 100 % dilution.

DES	Dynamic Viscosity (kg m^{-1} s^{-1})
DES-1 (BTEAC/RES)	3.6895
DES-2 (BTEAC/CH)	3.8087
DES-3 (BTEAC/PH)	3.5591
DES-4 (BTEAC/PY)	4.3415

Report error in viscosity-2.576.

four. According to coumarin-102, DES-4 is a lot more dipolar than DES-2 and DES-1. The findings demonstrate the polarity of the four solvents in the following order: DES-4 > DES-1ˆ > DES-2 > DES-3. The probe-reported dipolarity of the DES is unquestionably controlled by the H-bond donor that was utilized to produce it. The results of these investigations will provide fresh opportunities for DES applications in the chemical and material sciences (see Fig. 11).

6.3. Solvatochromic parameters of synthesized DESs

The Kamlet Taft (KT) parameters, which furnishes the information between several "polarity" properties, include acidity (α), basicity (β), and dipolarity/polarizability (π*) of solvents give valuable information about cybotactic region. The KT properties of the solvent can be used to correlate and predict the kinetic rate constants in different solvents. Further, the solvatochromic parameters gives important information regarding solute–solvent and solvent–solvent interaction. The following

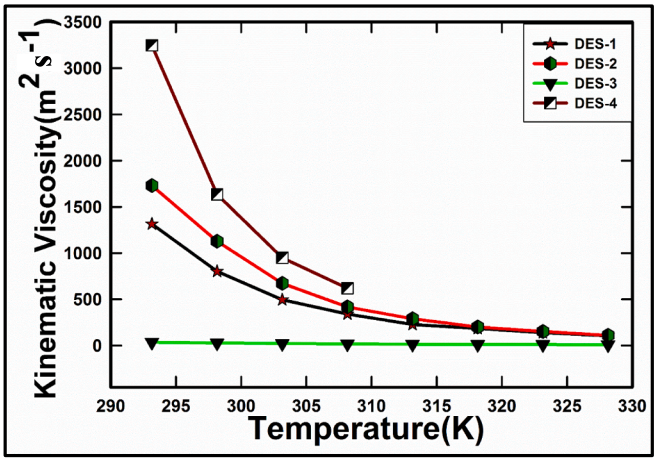


Fig. 9. Variation of kinematic viscosity for four pure DESs as a function of temperature (290–350) K.

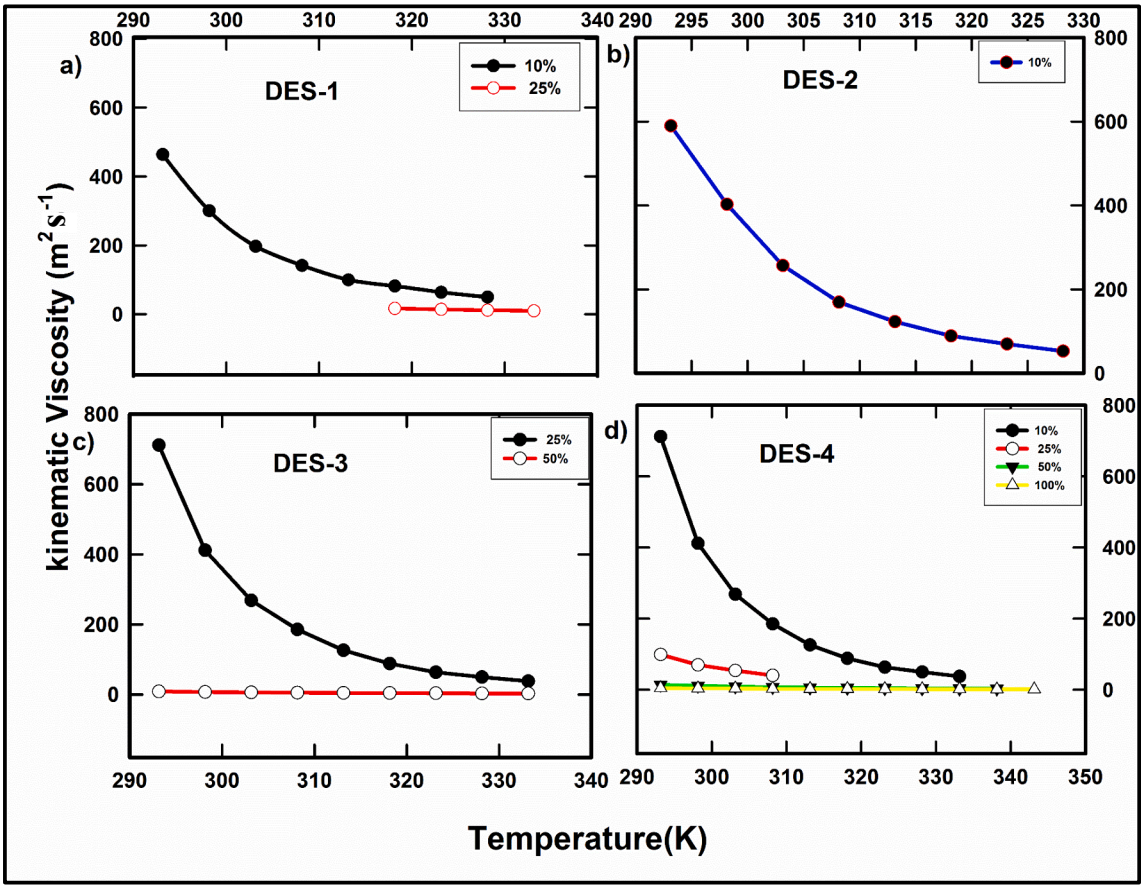


Fig. 8. Variation of kinematic viscosity for four synthesized DESs + water mixture (10, 25, 50 100 wt% dilution) as a function of temperature (290–350 K) at 101.13 kPa.

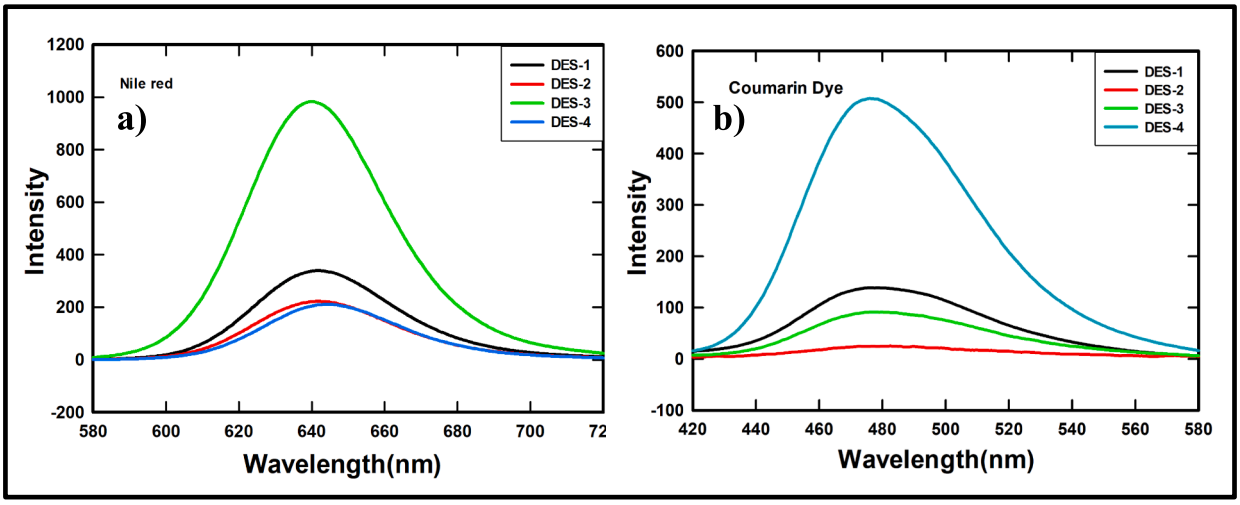


Fig. 10. (a) Fluorescence emission spectra of Nile red (NR, 10 mM, λ_{ex} = 550) and (b) Coumarin (CRM, 10 mM, λ_{ex} = 365 nm;) in four DESs under ambient conditions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

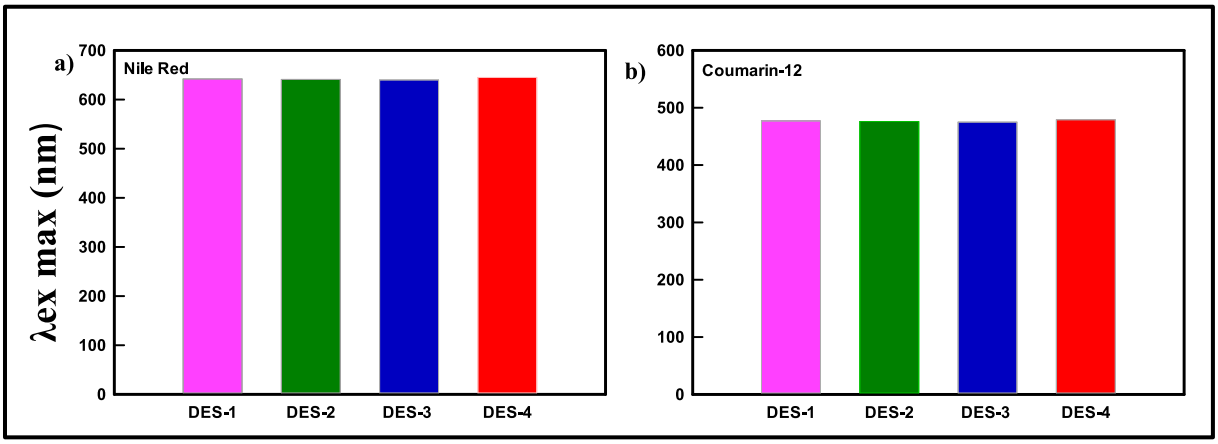


Fig. 11. Bar graph of fluorescence emission maxima (λ_{ex} max) of (a) Nile red and (b) coumarin12 dissolved in four DESs, under ambient conditions. For Nile red, λ_{ex} = 550 nm; (DES-1-642, DES-2-641, DES-3-640, and DES-4-645) and for coumarin-12, λ_{ex} 365 nm (DES-1-477, DES-2-476, DES-3-475, and DES-4-479). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

equations are used for the determination of KT parameters of DESs by using solvatochromic probes.

$$E_{T(NR)}\text{kcal.mol}^{-1} = 28591.44/\lambda_{MAX}(\text{nm}) \quad (1)$$

$$\alpha = (19.9657 - 1.0241\pi^* - \overset{\text{Å}}{\upsilon}_{NR}\text{max})/1.6078 \quad (2)$$

where $\overset{\text{Å}}{\upsilon} = 1/(\lambda_{MAX}{}^{*}10^{-4})$

$$\pi^* = -0.376{}^{*}\overset{\text{Å}}{\upsilon}_{NR}\text{max} + 7.73 \quad (3)$$

The Kamlet–Taft parameters dipolarity/polarizability (π*) and the hydrogen bonding acidity (α) of DESs were estimated using Nile Red solvatochromic probes and eqs −2 and 3. The (π*) evaluates non-specific interactions such as dipole–dipole and dipole-induced dipole interactions as well as the polarizability between the dye and the DESs. The results show that nonspecific interactions are employed as an index for DES polarity and that their value diminishes with increasing HBD OH group in polyols [61]. The very acidic nature of the DESs required the use of Nile Red to determine ET values. The ET(30) dye becomes protonated and loses its absorption band in an acidic solution, rendering it unfit as a probe for the solute–solvent interaction. Because of its low basicity, photochemical stability, solubility in a range of solvents, and other unique characteristics, Nile red was chosen as a solvatochromic probe in acidic DESs. Fig. 12 shows a bathochromic shift in the Nile red maximum as the chain length of the number of OH groups grew in DES. Nile red shows a bathochromic shift in polar solvents whereas a hypsochromic shift in nonpolar solvents [61]. Using eqs-3 and 4 polarity/polarizability (π*) and hydrogen bonding acidity (π) Kamlet-Taft parameters can be calculated. The π* evaluates non-specific interactions such as dipole–dipole and dipole-induced dipole interactions as well as the polarizability between the dye and the DESs.[88,92] The results show that nonspecific interactions are a measure of DES polarity and that they decrease with increasing HBD OH group size. As shown in Table 5, for example, the nonspecific interactions in DESs at 298 K decrease from DES-4 > DES-1 > DES-2 > DES-3, indicating that even a little change in the donor group has a considerable impact on the π* parameter. Additionally, it can be seen that hydrogen bond donors significantly contribute to the nonspecific contacts and solvent polarity These findings demonstrate that the OH group between ammonium-based salts and polyalcohols HBD causes the π * values to decrease. The ability of the solvent to give protons is described by the Kamlet–Taft hydrogen bonding acidity (α), which is derived from the $E_{T(NR)}$ and π* values (eqns 2 and 3). According to the findings, the hydrogen bonding acidity of DESs at 298 K ranges from 1.8159 to 2.0702. The presence of the OH

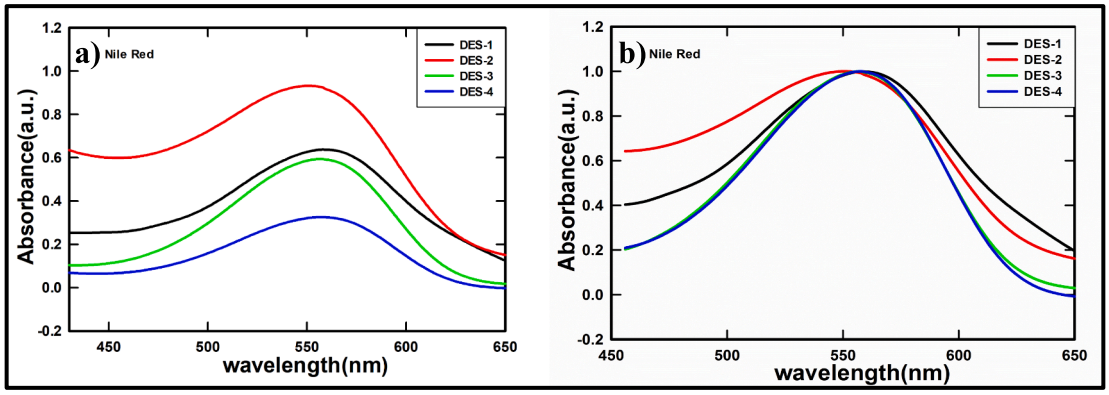


Fig. 12. (a) Nile red dye absorption spectra in dess (1:2) at 298 K, (b) normalized spectra of Nile red spectra, to get λ_{max} of DES. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Kamlet–Taft parameters of DESs at 298 K using above equation derived from Fig. S6.

DES	Max wave λmax	E_T parameter	ú NRmax	π*	α
DES-1 (BTEAC/ RES)	558	51.2391	17.921	0.99	2.0388
DES-2 (BTEAC/ CH)	557	51.3311	17.953	0.97	1.8857
DES-3 (BTEAC/ PH)	551	51.8892	18.148	0.90	1.8159
DES-4 (BTEAC/ PY)	559	51.14669	17.889	1.00	2.0702

group in the DESs may be the cause of their extreme acidity. The OH group's size or the HBD's molar ratio both significantly affect the parameter. It is worthy to mention here that Zhang et al reported a range of DES with different combinations of HBA and HBD components [25]. The Kamlet-Taft parameters of the prepared DES have been evaluated, which shows that the structure of the HBA and HBD components has a significant influence on the KT parameters. While the α value is slightly higher in comparison to that reported by Zhang et al. [25].

7. I_2 capture efficiency of dess: kinetics study

Initially, the extinction coefficient of I_2 has been determined by UV–Vis spectroscopic study (Figs. S7 and S8). The absorbance spectra of the separated hexane layer have been recorded over time with time intervals (1.5 min, 2.15 min, 3.30 min, 4.1 hrs, 5.2 hrs, 6.3 hrs, 7.4 hrs, 8.5 hrs, and 9.6 hrs). From, recorded spectra, conc. of I_2 in the hexane layer is determined. The extraction efficiency was calculated as:

$$\%ExtractionEfficiency = (Co - C_f/Co)*100 \quad (1)$$

where C_o and C_f represent the initial and final concentration, respectively. The Eq. (1) may be wrieen as

$$\%ExtractionEfficiency = (A_0 - A)/A_0 \quad (2)$$

(a) DES-0.5 g + Iodine hexane stock (2 mM)solution-3 ml

Fig. S9 shows that I2 capture tendency of all four prepared DESs

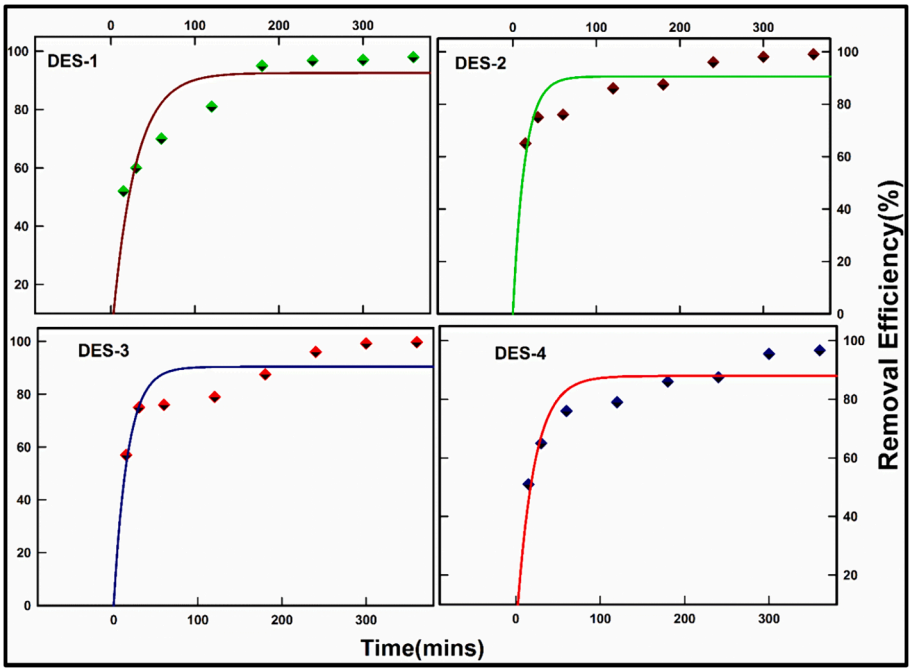


Fig. 13. Removal efficiency (%) of iodine by synthesized DESs and corresponding curve fitted by equation $Y = a(1 - exp - bx)$.

[45]. The efficiency for all DES has been calculated by UV–Vis spectroscopy. Fig.S9. shows visual evidence of the absorption of I_2 by DES. Further, efficiency is calculated by eqn (1) and given in Fig. 13. It is observed that the efficiency increases with time. The DESs' structure demonstrates that the more OH groups present, the more effectively they remove iodine. The enhanced efficiency of radioactive iodine (I2) capture in synthesized deep eutectic solvent (DES) can be attributed to a specific mechanism involving the interplay of electrostatic and hydrogen bonding interactions. The presence of benzyl triethyl ammonium chloride in the DES introduces electrostatic attractions between chloride and ammonium ions, which are essential for solubilizing iodine molecules. However, the formation of hydrogen bonds within the DES matrix serves to weaken the electrostatic attraction between chloride and ammonium ions. This reduction in electrostatic forces allows for stronger interactions between chlorine and iodine molecules, facilitating more effective iodine removal. Consequently, the synergy between weakened electrostatic attractions and strengthened hydrogen bonding leads to improved efficiency in capturing radioactive iodine. Further, it is worthy to mention here that the DES with high α shows higher efficiency. The reason for this may be easily formation of the halogen bonding, however, the role of other interaction may not be ruled out.

(b) DES-1 g + Iodine hexane stock (2 mM)solution-3 ml

From Figs. S10 and S11. of UV spectra of iodine-hexane stock with DESs, we got the maximum absorbance wavelength. From Figs. 13 and 14, it is clearly shown after passing time removal efficiency of DESs increases. The rate and capacity of iodine removal are changed when the mass of DESs is altered. The color of the iodine solution captured by DES-2 becomes violet into nearly white within 1 hr, and DES-3 takes two hrs; however, the solution is highly violet after 3–4 hrs in DES-1 and DES-4 and takes almost 7hrs to change it to white. From Table 6. R^2 of the fitted curves for DES-1, DES-2, DES-2, and DES-4 are 0.6335, 0.6110, 0.6831 and 0.8224, respectively. Suggesting a favorable regression with the model $Y = a(1 - exp - bx)$. Comparing DES(0.5gm) data and DES (1 gm) we also conclude that increasing the DES ratio also increases the removal efficiency % of iodine from the stock solution. From above Tables 6 and 7, we can make a comparative study of the efficiencies of different deep eutectic solvents for the removal of iodine from hexane solution. It is clear from the data that the removal efficiency decreases with an increase in the number of OH group of DESs means with an increase in the acidic nature the removal efficiency decreases. As we conclude, the removal efficiency trend in four novel DESs is – DES-3 > DES-1 > DES-2 > DES-4. Further, removal of I_2 from the iodine hexane layer follows the kinetics given by the equation $Y = a(1 - exp - bx)$. The fitting was done for data and the recovered constant are given in Tables 6 and 7. Which shows Parameter *b* (Iodine Capture Efficiency): In Table 7. The values of (a) range from approximately 94.7993 to 97.3008, with a mean value of around 96.12445. While in Table 6. The values of (*a*) range from approximately 87.9383 to 92.5651, with a mean value of around 90.856375. Considering the difference in the amount of DES used, it appears that when 1 g of DES is used (Table 7), the efficiency of iodine capture tends to be higher compared to when 0.5 g of DES is used (Table 6). This suggests that increasing the amount of DES used might lead to higher efficiency in capturing iodine, assuming other factors remain constant. Other Parameters (b and R^2) in Table 6. The values of parameter *b* and the coefficient of determination (R^2) are generally higher compared to Table 7. This might indicate that while increasing the amount of DES used improves iodine capture efficiency (*a*), it might not necessarily improve the overall fit of the model (as indicated by *b* and R^2). So the comparison suggests that increasing the amount of DES used might enhance iodine capture efficiency, but it could also affect other aspects of the relationship described by the model. It is worthy to mention here some recent findings about I2 capture using DES and ILs. Yan et al. [62] reported the removal of I2 by ILs

Table 6
Shows the removal efficiency (%) of iodine at room temperature, values derived by the curve fitting of Fig. 13 by following equation $Y = a(1 - exp - bx)$.

DES	a	b	Rsqr
DES-1 (BTEAC/RES)	92.5651	0.0365	0.6335
DES-2 (BTEAC/CH)	90.5451	0.0713	0.6110
DES-3 (BTEAC/PH)	90.3770	0.0596	0.6831
DES-4 (BTEAC/PY)	87.9383	0.0481	0.8224

Table 7
Shows the removal efficiency (%) of iodine at room temperature, values derived by the curve fitting of Fig. 14. by following equation $Y = a(1 - exp - bx)$.

DES	a	b	Rsqr
DES-1 (BTEAC/RES)	96.6377	0.1681	0.6133
DES-2 (BTEAC/CH)	95.7600	0.5230	0.2890
DES-3 (BTEAC/PH)	97.3008	0.6085	0.4785
DES-4 (BTEAC/PY)	94.7993	0.4979	0.5665

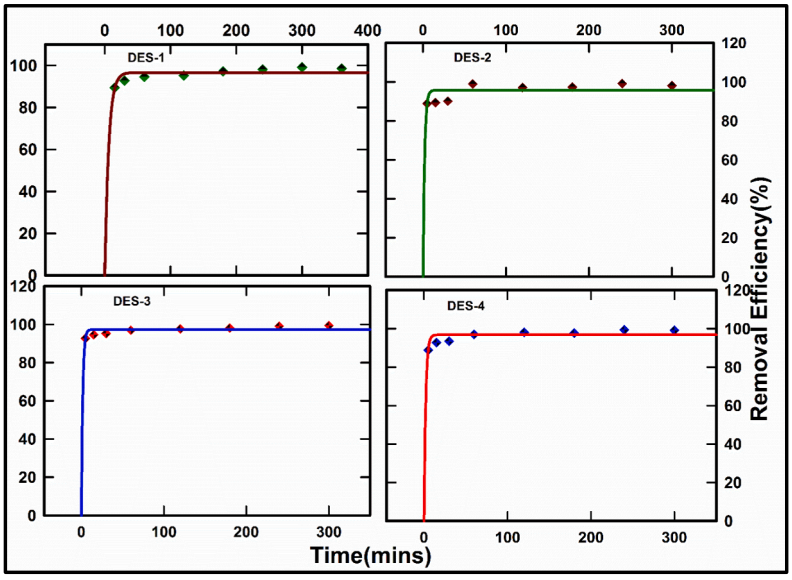


Fig. 14. Removal efficiency (%) of iodine by synthesized DESs and corresponding curve fitted by equation $Y = a(1 - exp - bx)$.

which shows anions dependence while cations of ILs have little influence. It is observed that halogen bonding plays a vital role here . Similarly, Li et al. [63] reported the removal of I2 with low cost DES . In this case also role of halogen bonding between I2 and DES has been cited. Chen et al. [46] reported I2 capture by amino acid and PEG based DES and explore effect of several factors (molar ratio, mass of DES phase, and volume of I2 phase) on I2 capture [44,45].

8. Conclusion

The results of this study show that DESs are potential future materials for extensive industrial radioiodine removal. In this study, the preparation of four BTEAC-based DESs and the study of their physical properties such as water solubility, density, viscosity, and FTIR and ^{1}HNMR spectroscopic analysis shows the interaction of BTEAC and aromatic phenols show I_2. The ability of the deep eutectic solvent to capture iodine from hexane solutions was then tested. The outcomes demonstrated that the DESs were effective at capturing iodine. The maximum iodine uptake of 80 % at a solvent-to-iodine ratio of 1:2. Further was observed DES polarity HBD and HBA interaction and solvatochromism parameters are studied with fluorescence spectroscopy and UV–Vis spectroscopy with probes Nile red and coumarin. All these results give a conclusion that the more polar the DES is the less efficient for the removal of iodine, DES-4 is the most polar and least efficient for iodine removal, and follow the trend by DES-4 < DES-2–DES-1 < DES-3. The efficiency of I_2 capture could range from 85 % to 99 % along with the change of mass ratio of DES-3. Overall, the findings indicate that the deep eutectic solvent synthesized in this study has the potential to be a new material for iodine capture. The DESs have several advantages over other solvents and adsorbents, including low volatility, good thermal stability, and a high iodine uptake capacity [59]. However, additional research is required to optimize the solvent composition, assess its stability under different conditions, and evaluate its performance in real-world applications [26,28,29,41,64]. Finally, the successful synthesis and characterization of the deep eutectic solvent, as well as its effective application for iodine capture, represent a significant progression in the development of new materials for environmental remediation. Interest in DESs is rapidly growing due to their disruptive potential and diverse applications. Both startups and established corporations worldwide are increasingly recognizing the value of DES technology. Fostering innovation and collaboration is crucial for translating DES research into commercial products. By supporting entrepreneurial efforts and technology transfer, we can speed up DES adoption. Scalability is key, as practical use depends on cost-effective, large-scale production. While initial experiments show DESs' effectiveness in iodine capture, achieving commercial potential requires scalable manufacturing, optimized production, and economic feasibility. Moreover, our exploration of DES applications extends far beyond iodine capture, encompassing a diverse array of industries and scientific disciplines [27,32,33].

CRediT authorship contribution statement

Masooma Siddiqui: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Md Sayem Alam:** Writing – review & editing, Software, Data curation, Conceptualization. **Maroof Ali:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2024.125883.

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